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To:

The Editor-in-Chief
Expert Systems with Applications (ESWA)
Elsevier Science

Subject: Submission of Original Research Manuscript for Double-Blind Peer Review

Dear Editor-in-Chief,

We are pleased to submit our original research manuscript entitled:

“Risk-Controlled Spatio-Temporal Graph Learning for Anti-Money Laundering Under Extreme Imbalance and Topological Camouflage”

for consideration as a regular research paper in *Expert Systems with Applications*.

Scope and Alignment with ESWA Mandate:

In direct alignment with the core mandate of *Expert Systems with Applications*, this work develops and validates an end-to-end intelligent decision-support and surveillance expert system—named **C-STGB** (Conformal Spatio-Temporal GraphBoost)—specifically engineered for the critical domain of financial forensics and anti-money laundering (AML). We confirm that this work contains **no military or defense applications**. Rather than presenting an abstract machine learning heuristic or nature-inspired metaphor, C-STGB tackles five foundational domain hurdles in banking compliance: continuous-time velocity evasion, adversarial topological camouflage, extreme class imbalance ($\pi < 0.05\%$), cold-start account sparsity, and uncalibrated decision uncertainty.

Key Contributions and Substantive Findings:

- **Continuous-Time Multi-Scale Temporal Graph Architecture:** C-STGB unifies continuous harmonic temporal embeddings, Hawkes arrival intensity, and a context-aware edge-trust filter ($\hat{g}_{ij}$) that prunes 65.9% of camouflage chaff connecting to high-degree banking hubs.
- **Flow-Conserving Typology Oversampling:** A latent GraphSMOTE mechanism clustered by laundering archetypes with Asymmetric Focal Tversky optimization, formally proven to preserve point-level mass-flow conservation invariants.
- **Finite-Sample Risk Guarantees under Drift:** Class-Conditional Conformal Risk Control establishing $\geq 99.0\%$ empirical true-label coverage under within-class exchangeability, non-asymptotic drift bounds, and online Martingale tracking via Adaptive Conformal Inference (ACI). Decisive singletons route $> 99.4\%$ of volume through automated straight-through tiers, reducing officer review queues to 0.50%.
- **Large-Scale Empirical Corpus & System Latency:** Evaluated across 14 financial transaction networks (9.53M entities, 9.53M transactions across 182 evaluation pairs over five locked seeds), C-STGB achieves 67.26% unweighted macro F1, delivering +43.49 pp mean uplift over deep spatial GNNs ($W = 105.0, p_{\text{adj}} < 0.001$). Streaming Fast-Path inference executes in 0.45 ms per single event (31, 200 tx/s).
- **Compliance Decision Support Prototype:** An auxiliary module assisting examiners in compiling verifiable FinCEN Form 111 XML dossiers under Federal Reserve SR 11-7 model risk governance standards.

Submission Declarations & Policy Compliance:

- **Double-Blind Peer Review:** In strict compliance with ESWA policy, two separate files are provided: (1) an unanonymized Title Page with all author details, affiliations, ORCIDs, CRediT statement, and acknowledgments; and (2) a completely blinded manuscript with zero identifying information and anonymized repository links.
- **Research Data Policy (Option C):** All 14 benchmark datasets are publicly accessible. Complete pipelines, configuration files, and reproduction scripts have been deposited in an anonymized repository (<https://anonymous.4open.science/r/Intelligent-AML>) and will be archived with a permanent Zenodo DOI upon publication.
- **Declaration of Generative AI:** Disclosed in a dedicated mandatory section preceding the references.
- **Originality:** This manuscript is original, has not been published previously, and is not under review elsewhere.

Thank you for your consideration of our manuscript. We look forward to receiving the reviewers' constructive feedback.

Sincerely yours,

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